



Audit Event Chain with Merkle Checkpoints

Public Technical Disclosure – Prior Art Publication

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Abstract

This document discloses an append-only audit log architecture using **hash chaining for tamper evidence** and **periodic Merkle tree aggregation** for efficient verification. The system supports **optional blockchain anchoring** across any blockchain that accepts commitment data, making it chain-agnostic by design. This preserves a minimal on-chain footprint—only compact commitment roots are anchored, not full logs.^{[1][2]}

This specification is published as prior art to establish the hash-chained audit log with Merkle checkpoint pattern and prevent patent enclosure of blockchain-agnostic audit trail methods.

1. Problem Statement

Autonomous agent systems require audit trails that satisfy four properties:

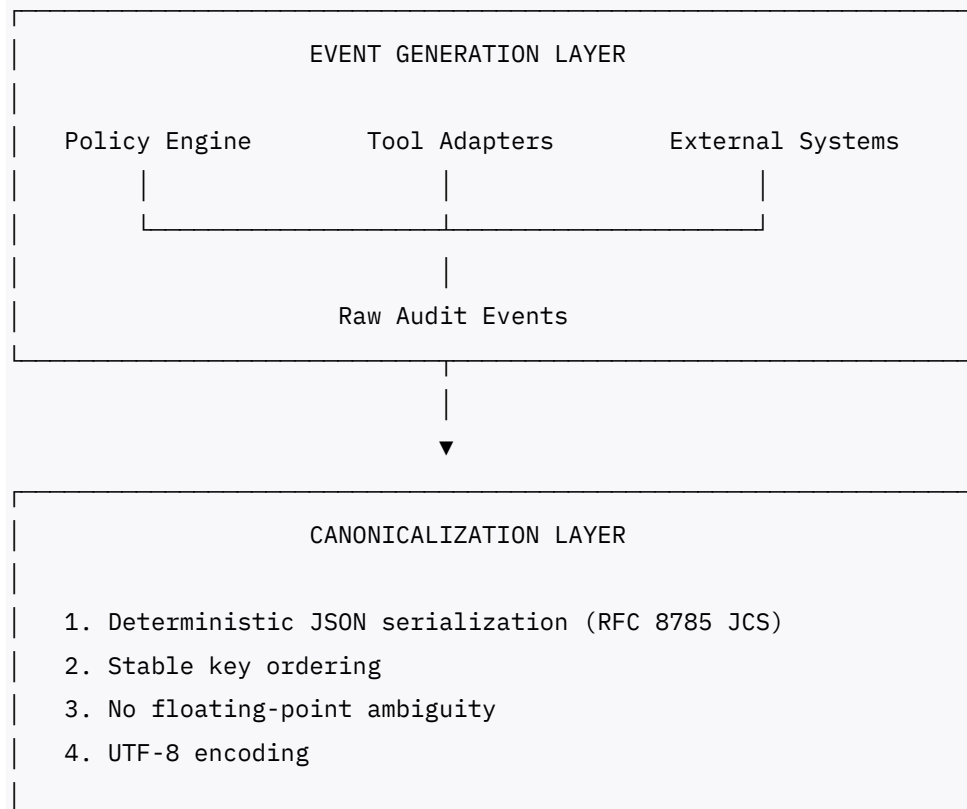
1. **Tamper-evidence:** Any modification to historical records is detectable
2. **Efficient verification:** Individual events can be verified without scanning entire logs
3. **Chain-agnostic anchoring:** Not locked to any single blockchain
4. **Privacy-preserving:** No PII in audit records; only commitments and metadata

Traditional approaches fail because:

- Centralized logs can be modified by operators
- Full log replication to blockchain is prohibitively expensive
- Single-chain anchoring creates vendor lock-in
- Storing raw data exposes sensitive information

The Solution: Append-only hash chains with periodic Merkle checkpoints, anchored to multiple blockchains by commitment only.

2. Core Architecture



Output: Canonical event bytes



HASH CHAIN LAYER

$\text{Event}[N].\text{prev_event_hash} = \text{Event}[N-1].\text{event_hash}$

$\text{Event}[N].\text{event_hash} = \text{Hash}(\text{canonical_bytes} + \text{prev_event_hash})$

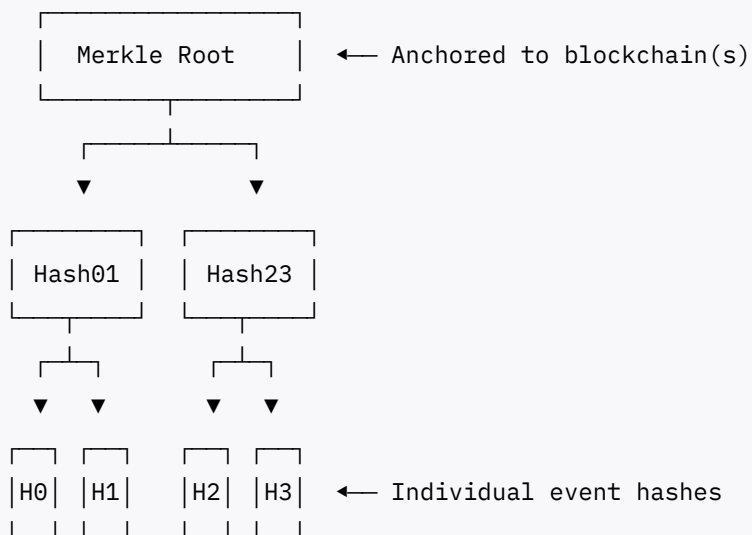


Tamper-evidence: Modifying any event invalidates entire chain

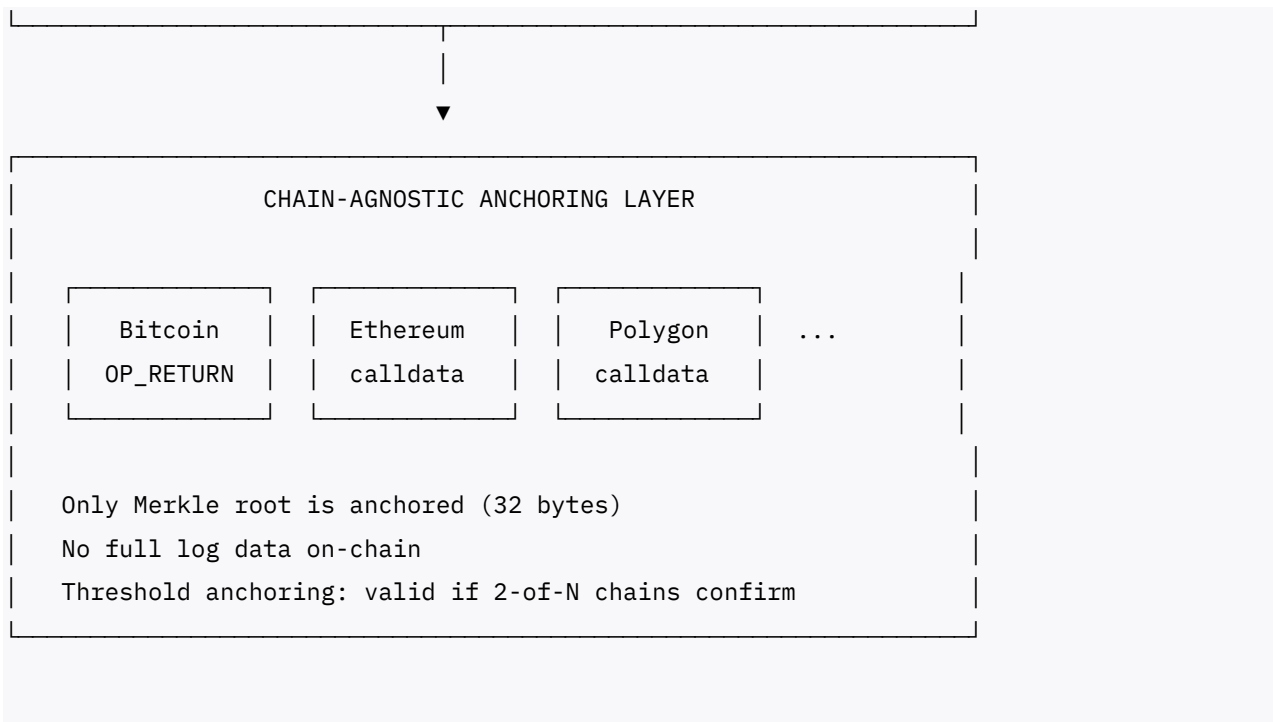


MERKLE CHECKPOINT LAYER

Every N events (or T minutes):



Efficient verification: $O(\log N)$ proof for any event



3. Event Structure

3.1 Audit Event Schema

Every enforcement decision, policy evaluation, or state transition produces an audit event:^{[2][1]}

```
interface AuditEvent {  
  // Event identity  
  event_id: string;           // Unique identifier (e.g., "evt_01H...")  
  event_type: EventType;     // Category of event  
  timestamp: number;         // Unix timestamp (milliseconds)  
  prime_index: number;       // Prime epoch at event time  
  
  // Actor context  
  actor: {  
    actor_type: "service" | "user" | "agent";  
    actor_id: string;         // Policy engine, adapter, etc.  
  };  
  
  // Correlation (for tracing)  
  correlation: {
```

```

    request_id: string;
    session_id: string;
    subject_id: string;           // Hash only, never raw
    intent_id: string;
};

// Cryptographic bindings (hashes only)
bindings: {
    policy_bundle_hash: string;
    intent_hash: string;
    token_id?: string;
    ltrace_hash?: string;        //  $\Lambda$ -Trace decision reference
};

// Event-specific details
details: Record<string, unknown>;

// Privacy controls
privacy: {
    redactions: string[];        // Fields redacted
    sealed_refs?: string[];      // References to sealed content
};

// Hash chain integrity
integrity: {
    prev_event_hash: string;      // Previous event's hash
    event_hash: string;          // This event's hash
};

// Optional signature
signature?: string;              // Actor's signature over event
}

type EventType =
    | "INTENT_PROPOSED"
    | "POLICY_EVALUATED"
    | "CAPABILITY_ISSUED"
    | "CAPABILITY_REVOKED"
    | "EXECUTION_ATTEMPTED"
    | "EXECUTION_COMPLETED"

```

```
| "EXECUTION_BLOCKED"  
| "EXECUTION_SILENT"  
| "PROOF_VERIFIED"  
| "POLICY_BUNDLE_UPDATED"  
| "PRIME_GATE_ACTIVATION"  
| "XI_STATE_TRANSITION"  
| "IDENTITY_REGISTERED"  
| "AUDIT_CHECKPOINT_CREATED";
```

3.2 Privacy Guarantee

Audit events contain only hashed commitments and operational metadata:^[2]

INCLUDED IN EVENTS:

- ✓ Hash of intent (not raw params)
- ✓ Hash of subject identity (not raw user ID)
- ✓ Hash of policy bundle (not raw rules)
- ✓ Token ID (opaque identifier)
- ✓ Timestamps and durations
- ✓ Decision outcome (ALLOW/DENY/SILENT)
- ✓ Rule ID that matched
- ✓ Error codes

EXCLUDED FROM EVENTS:

- ✗ Email addresses, phone numbers
- ✗ Raw intent parameters (URLs, amounts, recipients)
- ✗ IP addresses
- ✗ User-agent strings
- ✗ Raw PII of any kind
- ✗ Wallet addresses (if sensitive)
- ✗ Query text, code snippets

4. Hash Chaining Algorithm

4.1 Event Hash Computation

ALGORITHM: ComputeEventHash(event, prev_hash)

INPUT:

event - The audit event object (without event_hash field)
prev_hash - Hash of the previous event (or GENESIS_HASH for first event)

OUTPUT:

event_hash - 32-byte hash uniquely identifying this event

STEPS:

1. Set event.integrity.prev_event_hash = prev_hash
2. Remove event.integrity.event_hash field (if present)
3. Serialize event to canonical JSON (RFC 8785 JCS)
4. Encode canonical JSON to UTF-8 bytes
5. event_hash = SHA256(canonical_bytes)
6. Return event_hash

4.2 Chain Integrity Invariant

FOR ALL events $E[i]$ where $i > 0$:

$E[i].\text{integrity.prev_event_hash} == E[i-1].\text{integrity.event_hash}$

GENESIS EVENT $E[0]$:

$E[0].\text{integrity.prev_event_hash} == \text{GENESIS_HASH}$

where $\text{GENESIS_HASH} = \text{SHA256}(\text{"LAMBDA_PROOF_AUDIT_CHAIN_GENESIS_V1"})$

4.3 Tamper Detection

ALGORITHM: VerifyChainIntegrity(events[])

INPUT:

events[] - Array of audit events in chronological order

OUTPUT:

valid - Boolean indicating chain integrity
broken - Index of first tampered event (if any)

STEPS:

```
1. If events[^0].integrity.prev_event_hash != GENESIS_HASH:
    Return (false, 0)

2. For i = 0 to len(events) - 1:
    a. recomputed = ComputeEventHash(events[i], events[i].integrity.prev_event_hash)
    b. If recomputed != events[i].integrity.event_hash:
        Return (false, i)
    c. If i > 0 AND events[i].integrity.prev_event_hash !=
events[i-1].integrity.event_hash:
        Return (false, i)

3. Return (true, -1)
```

5. Merkle Checkpoint Protocol

5.1 Checkpoint Structure

```
interface AuditCheckpoint {
    checkpoint_id: string;           // Unique identifier
    checkpoint_type: "PERIODIC" | "MANUAL" | "THRESHOLD";

    // Coverage
    event_range: {
        from_event_id: string;
        to_event_id: string;
        event_count: number;
    };

    // Merkle tree
    merkle_root: {
        algorithm: "sha256";
        value: string;               // 32-byte hex
    };

    // Tree structure (for verification)
    tree_depth: number;
    leaf_count: number;
```



```

// Timing
created_at: number;
prime_index: number;

// Anchoring status
anchoring: {
  mode: "NONE" | "SINGLE" | "THRESHOLD";
  threshold?: number;          // e.g., 2-of-3
  anchors: AnchorRecord[];
};
}

interface AnchorRecord {
  chain: string;                // "bitcoin", "ethereum", "polygon", etc.
  chain_id?: number;            // EVM chain ID
  method: "OP_RETURN" | "CALLDATA" | "TIMESTAMP_SERVICE";
  tx_hash?: string;             // Transaction hash (once confirmed)
  block_number?: number;
  confirmation_status: "PENDING" | "CONFIRMED" | "FINALIZED";
  timestamp?: number;
}

```

5.2 Merkle Tree Construction

ALGORITHM: BuildMerkleTree(event_hashes[])

INPUT:

event_hashes[] - Array of event hashes to include in tree

OUTPUT:

merkle_root - Root hash of the Merkle tree
 tree - Full tree structure (for proof generation)

STEPS:

1. If len(event_hashes) == 0:
 Return (EMPTY_TREE_ROOT, empty_tree)
2. Pad event_hashes to next power of 2:

```

        while len(event_hashes) is not power of 2:
            event_hashes.append(EMPTY_LEAF_HASH)

3. Initialize leaves:
    leaves = event_hashes

4. Build tree bottom-up:
    current_level = leaves
    tree_levels = [current_level]

    while len(current_level) > 1:
        next_level = []
        for i = 0 to len(current_level) step 2:
            left = current_level[i]
            right = current_level[i+1]
            parent = SHA256(left + right) // Concatenate then hash
            next_level.append(parent)
        current_level = next_level
        tree_levels.append(current_level)

    merkle_root = current_level[0]

5. Return (merkle_root, tree_levels)

```

5.3 Merkle Proof Generation

ALGORITHM: GenerateMerkleProof(event_hash, tree)

INPUT:

event_hash - Hash of the event to prove inclusion
 tree - Full Merkle tree structure

OUTPUT:

proof - Array of sibling hashes for verification
 index - Leaf index of the event

STEPS:

1. Find leaf index:
 index = tree.leaves.indexOf(event_hash)

```

    If index == -1: Return ERROR("Event not in tree")

2. Build proof path:
    proof = []
    current_index = index

    for level = 0 to tree.depth - 1:
        sibling_index = current_index XOR 1 // Toggle last bit
        sibling_hash = tree[level][sibling_index]
        proof.append({
            hash: sibling_hash,
            position: "left" if sibling_index < current_index else "right"
        })
        current_index = current_index / 2 // Move to parent

3. Return (proof, index)

```

5.4 Merkle Proof Verification

ALGORITHM: VerifyMerkleProof(event_hash, proof, merkle_root)

INPUT:

```

event_hash - Hash of the event to verify
proof      - Array of sibling hashes
merkle_root - Expected root hash

```

OUTPUT:

```

valid      - Boolean indicating proof validity

```

STEPS:

```

1. current_hash = event_hash

2. For each step in proof:
    if step.position == "left":
        current_hash = SHA256(step.hash + current_hash)
    else:
        current_hash = SHA256(current_hash + step.hash)

```

```
3. Return current_hash == merkle_root
```

6. Chain-Agnostic Anchoring

6.1 Anchoring Strategies

The checkpoint Merkle root can be anchored to any blockchain that accepts data commitments:^[1]

Chain	Method	Data Size	Cost	Finality
Bitcoin	OP_RETURN	80 bytes max	~\$0.50-\$5	~60 min (6 blocks)
Ethereum	Calldata	Unlimited	~\$1-\$20	~15 min (finalized)
Polygon	Calldata	Unlimited	~\$0.01-\$0.10	~5 min
Arbitrum	Calldata	Unlimited	~\$0.05-\$0.50	~10 min
OpenTimestamps	Aggregated	32 bytes	Free	~60 min

6.2 Threshold Anchoring

For critical audit trails, use threshold anchoring:

THRESHOLD ANCHORING RULE:

A checkpoint is considered "anchored" when:

```
anchors.filter(a => a.confirmation_status == "FINALIZED").length >= threshold
```

Example: 2-of-3 anchoring

- Anchor to Bitcoin, Ethereum, and Polygon
- Checkpoint valid when any 2 chains confirm
- Survives single chain failure/reorg

6.3 Anchor Transaction Format

Bitcoin (OP_RETURN):

```
OP_RETURN <LAMBDA_PROOF_V1> <checkpoint_id[8 bytes]> <merkle_root[32 bytes]>
```

Ethereum/EVM (Calldata):

```
function anchorCheckpoint(  
    bytes32 checkpointId,  
    bytes32 merkleRoot,  
    uint256 eventCount,  
    uint256 fromTimestamp,  
    uint256 toTimestamp  
) external;
```

7. Standard Event Types

7.1 Intent Lifecycle Events

```
// A. INTENT_PROPOSED (Agent → Policy Engine)  
{  
    "event_type": "INTENT_PROPOSED",  
    "actor": {"actor_type": "agent", "actor_id": "orchestrator_v1"},  
    "details": {  
        "tool": "email",  
        "action": "send",  
        "risk_hint": "MEDIUM",  
        "params_commitment": "sha256:a1b2c3..."  
    }  
}  
  
// B. POLICY_EVALUATED (Policy Engine decision)  
{  
    "event_type": "POLICY_EVALUATED",  
    "actor": {"actor_type": "service", "actor_id": "policy_engine_v1"},  
    "details": {  
        "decision": "ALLOW",  
        "rule_id": "r2_email_send_requires_consent",  
        "proof_level": "ATTESTATION",  
        "required_evidence": ["user_consent_receipt"],
```

```

    "evidence_refs": ["evr_consent_001"]
  }
}

// C. CAPABILITY_ISSUED (Token minted)
{
  "event_type": "CAPABILITY_ISSUED",
  "actor": {"actor_type": "service", "actor_id": "policy_engine_v1"},
  "details": {
    "token_id": "cap_0d2b...",
    "expires_at": "2026-01-08T00:05:01Z",
    "audience": {"tool": "email", "adapter_id": "email_adapter_v1"},
    "scope_summary": ["send_to_commitment", "subject_commitment", "body_commitment"]
  }
}

```

7.2 Execution Events

```

// D. EXECUTION_ATTEMPTED (Adapter receives token)
{
  "event_type": "EXECUTION_ATTEMPTED",
  "actor": {"actor_type": "service", "actor_id": "email_adapter_v1"},
  "details": {
    "token_id": "cap_0d2b...",
    "verification": {
      "sig_ok": true,
      "not_expired": true,
      "intent_hash_match": true,
      "nonce_ok": true
    }
  }
}

// E. EXECUTION_COMPLETED (Action succeeded)
{
  "event_type": "EXECUTION_COMPLETED",
  "actor": {"actor_type": "service", "actor_id": "email_adapter_v1"},
  "details": {
    "result": "SUCCESS",

```

```

    "receipt": {
      "receipt_id": "rcpt_55c...",
      "receipt_hash": "sha256:d4e5f6..."
    },
    "external_ref": "smtp_message_id_opaque",
    "bytes_out": 10240,
    "duration_ms": 342
  }
}

// F. EXECUTION_BLOCKED (Verification failed)
{
  "event_type": "EXECUTION_BLOCKED",
  "actor": {"actor_type": "service", "actor_id": "payment_adapter_v1"},
  "details": {
    "reason": "INTENT_HASH_MISMATCH",
    "error_code": "ACI-005",
    "token_id": "cap_7e8f...",
    "expected_hash": "sha256:abc...",
    "received_hash": "sha256:def..."
  }
}

// G. EXECUTION_SILENT (CSL/Drift failure)
{
  "event_type": "EXECUTION_SILENT",
  "actor": {"actor_type": "service", "actor_id": "web_adapter_v1"},
  "details": {
    "reason": "DRIFT_EXCEEDED",
    "drift_score": 0.42,
    "drift_threshold": 0.30,
    "archived": true
  }
}

```

7.3 System Events

```

// H. AUDIT_CHECKPOINT_CREATED
{

```

```

"event_type": "AUDIT_CHECKPOINT_CREATED",
"actor": {"actor_type": "service", "actor_id": "archivum_v1"},
"details": {
  "checkpoint_id": "chk_01H...",
  "merkle_root": "sha256:9a8b7c...",
  "event_count": 10000,
  "range": {
    "from_event_id": "evt_a...",
    "to_event_id": "evt_b..."
  },
  "anchoring": {
    "mode": "THRESHOLD",
    "threshold": 2,
    "anchors": [
      {"chain": "bitcoin", "method": "OP_RETURN", "confirmation_status": "PENDING"},
      {"chain": "ethereum", "method": "CALLDATA", "confirmation_status": "PENDING"},
      {"chain": "polygon", "method": "CALLDATA", "confirmation_status": "PENDING"}
    ]
  }
}
}

// I. POLICY_BUNDLE_UPDATED
{
  "event_type": "POLICY_BUNDLE_UPDATED",
  "actor": {"actor_type": "service", "actor_id": "policy_admin"},
  "details": {
    "old_bundle_hash": "sha256:111...",
    "new_bundle_hash": "sha256:222...",
    "change_summary": "payments: daily limit reduced to $100"
  }
}

```

8. On-Chain Event Schema

For events that require on-chain emission (Solidity):^[2]


```

// SPDX-License-Identifier: CC0-1.0

// Core state transition event
event XiStateTransition(
    bytes32 indexed previousState, // Poseidon hash of prior state
    bytes32 indexed newState,      // Poseidon hash of new state
    address indexed executor,      // Transaction sender (not identity)
    uint256 timestamp,            // Block timestamp
    bytes32 proofHash             // BLAKE3 hash of zk-SNARK proof
);

// Prime gate activation event
event PrimeGateActivation(
    uint256 indexed primeIndex,    // Activated prime (validated via Miller-Rabin)
    address indexed activator,     // Actor triggering activation
    bytes32 proofHash             // Proof hash for audit linkage
);

// Identity registration event
event IdentityRegistered(
    address indexed entity,        // Wallet address (transient)
    bytes32 identityHash          // Poseidon(primeIdHash) - permanent identity
);

//  $\Delta$ -Trace linkage event
event LTraceLinked(
    bytes32 indexed ltraceHash,    // Canonical decision hash
    address indexed actor,         // Actor making the linkage
    uint256 tokenId               // Associated token ID
);

// Audit checkpoint anchored
event AuditCheckpointAnchored(
    bytes32 indexed checkpointId,
    bytes32 indexed merkleRoot,
    uint256 eventCount,
    uint256 fromTimestamp,
    uint256 toTimestamp
);

```

9. Verification Algorithms

9.1 Single Event Verification

ALGORITHM: VerifySingleEvent(event, checkpoint, proof)

INPUT:

event - The audit event to verify
checkpoint - The checkpoint containing the Merkle root
proof - Merkle proof for the event

OUTPUT:

valid - Boolean indicating event authenticity

STEPS:

1. Recompute event hash:

```
expected_hash = ComputeEventHash(event, event.integrity.prev_event_hash)
If expected_hash != event.integrity.event_hash:
    Return false // Event has been modified
```

2. Verify Merkle inclusion:

```
merkle_valid = VerifyMerkleProof(
    event.integrity.event_hash,
    proof,
    checkpoint.merkle_root.value
)
If not merkle_valid:
    Return false // Event not in checkpoint
```

3. Verify checkpoint anchoring (optional):

```
If checkpoint.anchoring.mode != "NONE":
    confirmed_count = checkpoint.anchoring.anchors
        .filter(a => a.confirmation_status == "FINALIZED")
        .length
    If confirmed_count < checkpoint.anchoring.threshold:
        Return false // Insufficient blockchain confirmations
```

4. Return true

9.2 Chain Segment Verification

ALGORITHM: VerifyChainSegment(events[], start_checkpoint, end_checkpoint)

INPUT:

events[] - Contiguous array of audit events
start_checkpoint - Checkpoint before first event (or null for genesis)
end_checkpoint - Checkpoint after last event

OUTPUT:

valid - Boolean indicating segment integrity
errors - Array of verification errors

STEPS:

1. Verify hash chain integrity:
 (chain_valid, broken_index) = VerifyChainIntegrity(events)
 If not chain_valid:
 Return (false, ["Chain broken at index " + broken_index])
2. Verify start boundary:
 If start_checkpoint != null:
 first_event = events[^0]
 If first_event.integrity.prev_event_hash != events_before_checkpoint_last_hash:
 Return (false, ["Gap between checkpoint and segment start"])
3. Verify end boundary:
 last_event = events[len(events) - 1]
 (last_in_tree, _) = GenerateMerkleProof(
 last_event.integrity.event_hash,
 end_checkpoint.merkle_tree
)
 If not last_in_tree:
 Return (false, ["Last event not in checkpoint"])
4. Verify all events in checkpoint:
 For each event in events:
 If event in end_checkpoint.event_range:

```
(in_tree, proof) = GenerateMerkleProof(event.event_hash, end_checkpoint.tree)
If not in_tree:
    Return (false, ["Event " + event.event_id + " missing from checkpoint"])
```

```
5. Return (true, [])
```

10. Audit Log File Format

Events are appended to persistent storage in newline-delimited JSON format:^[2]

```
{"timestamp":"2026-01-08T18:32:15Z","event_type":"INTENT_PROPOSED","event_id":"evt_001",
..., "integrity":{"prev_event_hash":"GENESIS","event_hash":"abc123..."}}
{"timestamp":"2026-01-08T18:32:16Z","event_type":"POLICY_EVALUATED","event_id":"evt_002"
,..., "integrity":{"prev_event_hash":"abc123...", "event_hash":"def456..."}}
{"timestamp":"2026-01-08T18:32:17Z","event_type":"CAPABILITY_ISSUED","event_id":"evt_003"
,..., "integrity":{"prev_event_hash":"def456...", "event_hash":"ghi789..."}}
```

Properties:

- One event per line (newline-delimited JSON / NDJSON)
- Append-only (no in-place modifications)
- Hash chain maintained across file rotations
- Checkpoints reference event IDs, not file offsets

11. Security Properties

Property	Mechanism	Guarantee
Tamper-evidence	Hash chaining	Modifying any event changes all subsequent hashes
Non-repudiation	Actor signatures	Events signed by generating service
Efficient verification	Merkle proofs	$O(\log N)$ proof for any event

Chain-agnosticism	Multi-chain anchoring	Not dependent on any single blockchain
Privacy preservation	Hash-only semantics	No PII in audit records
Temporal ordering	Monotonic timestamps + chain	Events cannot be reordered

12. Comparison with Related Work

System	Approach	Key Difference
This Specification	Hash chain + Merkle checkpoints + multi-chain anchoring	Chain-agnostic, privacy-preserving, efficient proofs
Certificate Transparency	Merkle trees + signed tree heads	Single-purpose (TLS certs), requires trusted logs
Hyperledger Fabric	Channel-based ledger	Requires permissioned network, heavy infrastructure
Ethereum Event Logs	On-chain events	Expensive, full data on-chain, single chain
OpenTimestamps	Bitcoin anchoring	Timestamp only, no event structure
Amazon QLDB	Hash-chained journal	Proprietary, single vendor

Key Novelty:

The combination of **append-only hash chaining** + **periodic Merkle aggregation** + **threshold multi-chain anchoring** + **hash-only privacy semantics** creates a verifiable audit trail that is portable across blockchains and preserves user privacy.

13. Conformance Requirements

13.1 Implementations MUST:

1. Compute event hashes using canonical JSON serialization (RFC 8785 JCS or equivalent)
2. Maintain hash chain integrity (each event references previous event's hash)

3. Support Merkle proof generation and verification
4. Emit `AUDIT_CHECKPOINT_CREATED` events for all checkpoints
5. Never store raw PII in audit events
6. Use append-only storage (no in-place modifications)

13.2 Implementations SHOULD:

- Support multiple blockchain anchoring targets
- Implement threshold anchoring for critical audit trails
- Provide APIs for Merkle proof generation
- Support checkpoint verification against on-chain anchors

13.3 Implementations MUST NOT:

- Modify historical events (append-only invariant)
 - Store raw sensitive data (emails, amounts, addresses) in events
 - Rely on a single blockchain for anchoring (vendor lock-in)
 - Break hash chain continuity across log rotations
-

14. Prior Art Statement

This specification is published as prior art under **CC0 1.0 Universal Public Domain Dedication** to:

1. **Establish prior art** for the hash-chained audit log with Merkle checkpoint pattern
2. **Prevent patent enclosure** of blockchain-agnostic audit trail methods
3. **Enable interoperability** across agent authorization systems
4. **Block trivial patent claims** on hash chaining, Merkle trees, and multi-chain anchoring for audit logs

What this document covers (PUBLIC):

- Event schema and structure

- Hash chaining algorithm
- Merkle tree construction and proof generation
- Chain-agnostic anchoring protocol
- Verification algorithms
- Standard event types

What remains proprietary (NOT DISCLOSED):

- Checkpoint frequency optimization algorithms
- Anchoring cost optimization strategies
- Event batching and compression techniques
- Replay cache implementation details
- Log rotation and archival strategies

15. Revision History

Version	Date	Changes
1.0	January 8, 2026	Initial specification

End of Specification

All four defensive publications are now complete:

1.  **Intent Canonicalization & Hashing Protocol**
2.  **Capability Token Binding Protocol**
3.  **Adapter Non-Bypassability Architecture**
4.  **Audit Event Chain with Merkle Checkpoints**

Would you like me to:

- **Consolidate all four** into a single document with a unified table of contents?

- **Create reference implementation pseudocode** for all four specs?
- **Generate test vectors** for conformance testing?
- **Draft the arXiv submission abstract** covering all four publications?

✱

-
1. lets-discuss-licensing-open-so-HV_bbj5uQlOPuXCF7b61YQ.md
 2. Audit-Logs_-Event-Schema-Sanitized-Examples.pdf
 3. Adapter Coordination Interface Specification.pdf
 4. Λ Proof Intent Schema Specification.pdf
 5. Conscious Sovereignty Layer (CSL) Solver.pdf
 6. Λ Proof Protocols.pdf
 7. DAO Charter.pdf
 8. Building a Λ Dapp_ A Developer's Implementation Guide.pdf
 9. Capability Token Semantic Mapping.pdf
 10. Data Sovereignty Informational Personhood.pdf
 11. Λ Proof_ An Architectural Blueprint for Verifiable Artificial Intelligence.pdf
 12. Λ^P -Archivum.pdf
 13. Λ Proof - R96 Credits.pdf
 14. Governance for a Non-Surveillance Digital Ecosystem.pdf
 15. Token Specification.pdf
 16. Risk Classification.pdf
 17. Policy Rules Specification.pdf
 18. Evidence Validation.pdf

19. How Trust is Measured.pdf

20. Λ Proof Token Issuance.pdf